

Human-Readable Proof Outline for the Kasami APN Theorem (Odd Dimension)

RequestProject / Kasami APN

1 What is proved

The Lean theorem `kasami_is_apn` proves that the Kasami power map

$$f(x) = x^{d(k)}, \quad d(k) = 2^{2k} - 2^k + 1,$$

is APN on a field F of characteristic 2 under the assumptions:

- $|F| = 2^n$,
- n odd,
- $1 < k < n$,
- k odd,
- $\gcd(k, n) = 1$.

Among these assumptions, the odd-dimension condition (n odd) becomes essential at Layer 5, where it is used to obtain the coprimality input needed to construct the linking exponent e' .

Also, at Layer 6, the Gold-map bijectivity is used to cancel a $(2^k + 1)$ -power, which is only valid under the coprimality assumption $\gcd(2^k + 1, 2^n - 1) = 1$.

2 APN meaning and reduction strategy

For a function $f : F \rightarrow F$, APN means that for every nonzero a and every collision

$$f(x + a) + f(x) = f(y + a) + f(y),$$

the only possible outputs are $y = x$ or $y = x + a$.

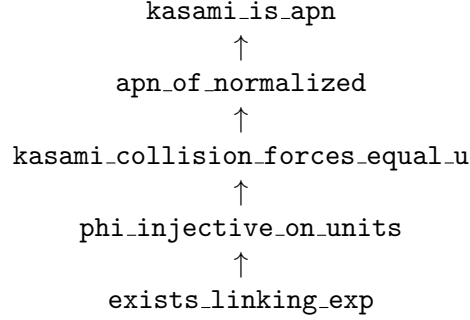
The proof does not attack this directly for all a . It first uses the normalization lemma `apn_of_normalized`: it is enough to prove the same uniqueness statement for the normalized equation with $a = 1$:

$$(x + 1)^d + x^d = (y + 1)^d + y^d \implies y = x \text{ or } y = x + 1.$$

So the whole theorem is reduced to controlling collisions of the normalized differential.

3 Layered proof architecture

The Lean file is organized in layers. The theorem-level dependency chain is:



Additional inputs are the Artin–Schreier identity, the key Kasami polynomial identity, Gold-map bijectivity, and a quadratic-kernel fact in characteristic 2.

Layer 1: Artin–Schreier trace identity

`truncTrace_artin_schreier` proves

$$L_k(x^2 + x) = x^{2^k} + x.$$

This is the basic conversion between truncated trace expressions and explicit powers.

Layer 2: Key Kasami identity

`kasami_key_identity` proves

$$((x+1)^d + x^d + 1)(x^2 + x)^{2^k} = (x^{2^k} + x)^{2^k+1}.$$

This is the bridge from differential terms to a form suited for the Φ -map analysis.

Layer 3: Gold coprimality and bijectivity

`gold_coprime` gives

$$\gcd(2^k + 1, 2^n - 1) = 1,$$

and `gold_pow_bijective` deduces that

$$y \mapsto y^{2^k+1}$$

is bijective on F .

This bijectivity is later used to cancel an outer $(2^k + 1)$ -power.

Layer 4: Arithmetic transfer identity

`kasami_arith_identity` is a modular arithmetic identity tailored to the exponent transfer step. It aligns the Kasami exponent arithmetic with the Dempwolff–Mueller bijection hypothesis.

Layer 5: Existence of a linking exponent e'

`exists_linking_exp` constructs e' with two modular constraints:

$$\begin{aligned} e'(2^k + 1) &\equiv 2^n - 1 - 2^k \pmod{2^n - 1}, \\ (2^{n-1} - 2^{k-1} - 1)e' &\equiv 2^{k-1} \pmod{2^n - 1}. \end{aligned}$$

Interpretation: e' is the arithmetic compatibility parameter that lets the map

$$\Phi(u) = \frac{L_k(u)^{2^k+1}}{u^{2^k}}$$

be rewritten as a composition of two bijective-style maps.

This is the first point in the proof where the hypothesis “ n odd” is genuinely needed: it enters through the Gold coprimality step used in constructing e' .

Layer 6: Injectivity of Φ on nonzero elements

`phi_injective_on_units` proves that for $u, v \neq 0$, a collision in the Φ -equation forces $u = v$. The proof combines:

- the exponent link from Layer 5,
- Gold-map bijectivity to cancel $(2^k + 1)$ powers,
- Dempwolff–Mueller theorem `LxXk'_bijective` to conclude equality of inputs.

The Gold-map bijectivity is only valid under the coprimality assumption $\gcd(2^k + 1, 2^n - 1) = 1$.

Layer 7: Differential collision implies equal Artin–Schreier values

`kasami_collision_forces_equal_u` proves:

$$(x + 1)^d + x^d = (y + 1)^d + y^d \implies x^2 + x = y^2 + y.$$

It applies Layer 2 to rewrite the collision, then:

- handles zero cases directly,
- uses Layer 6 injectivity in the nonzero case,
- uses Layer 1 for trace-power conversion.

This is the key bridge from normalized differential collisions to a quadratic relation in $x + y$.

Layer 8: Characteristic-2 kernel and normalization lemma

There are two lemmas used in the final step:

- `sq_add_self_eq_zero_char2`:

$$u^2 + u = 0 \iff u \in \{0, 1\}.$$

- `apn_of_normalized`: proving the normalized differential uniqueness implies full APN.

Layer 9: Final theorem

In `kasami_is_apn`, the final argument is short:

1. Apply `apn_of_normalized`; reduce to the $a = 1$ collision equation.
2. Use Layer 7 to get $x^2 + x = y^2 + y$.
3. Rearrange to

$$(x + y)^2 + (x + y) = 0.$$

4. Use Layer 8 kernel lemma to conclude $x + y = 0$ or $x + y = 1$.
5. Therefore $y = x$ or $y = x + 1$, exactly the APN condition.

4 One-paragraph summary

The proof is modular. It first reduces APN to a normalized differential equation, then converts that equation into a trace/exponent form via the Kasami identity, proves injectivity of the associated Φ -map on nonzero elements using a specially constructed linking exponent and known bijectivity theorems, and finally translates the resulting equality into the two-point alternative $y \in \{x, x + 1\}$ through a characteristic-2 kernel identity. This closes the APN theorem for Kasami exponents in odd dimension under the standard coprimality assumptions.

5 Literature review

5.1 Origin of the Kasami exponent

The exponent $d(k) = 2^{2k} - 2^k + 1$ first appeared in the coding-theoretic work of Kasami [1], who studied weight enumerators of subcodes of the second-order binary Reed–Muller codes. Kasami introduced the associated cyclic codes (now called Kasami codes) as a family of two-error-correcting codes meeting the Hamming bound for certain parameters. At that point the function was studied entirely for its coding properties; the connection to nonlinearity and cryptography came later.

5.2 APN functions: definition and cryptographic motivation

The concept of APN (Almost Perfect Nonlinear) function was introduced by Nyberg [2] to characterize S-box components of block ciphers that offer optimal resistance to the differential cryptanalysis attack of Biham and Shamir [3]. A function $f : \mathbb{F}_{2^n} \rightarrow \mathbb{F}_{2^n}$ is APN if for every nonzero a the equation $f(x + a) + f(x) = b$ has at most 2 solutions in x for every b . Power maps $x \mapsto x^d$ are the natural first family to study because their differential uniformity depends only on the exponent d modulo $2^n - 1$.

5.3 Gold functions: the first infinite APN family

The Gold exponent $2^k + 1$ had already appeared in Gold [4] in the context of maximal-length sequences and three-valued cross-correlation. Nyberg [2] reinterpreted this as the statement that $x \mapsto x^{2^k+1}$ is APN on $\mathbb{F}_{2^n}$ for every n satisfying $\gcd(k, n) = 1$. Gold functions are the simplest known APN power maps and their bijectivity ($y \mapsto y^{2^k+1}$ is a permutation under $\gcd(k, n) = 1$) is a key ingredient reused in the Kasami APN proof, where it cancels an outer $(2^k + 1)$ -power in the injectivity argument for Φ .

5.4 Known infinite families of APN power maps

A systematic search for APN power functions produced several infinite families. The currently known ones over $\mathbb{F}_{2^n}$ are:

1. **Gold** [4, 2]: $d = 2^k + 1$, $\gcd(k, n) = 1$. APN for all n .
2. **Kasami** [1, 5]: $d = 2^{2k} - 2^k + 1$, $\gcd(k, n) = 1$. APN for n odd (and more generally when $\gcd(3k, n) \in \{1, 3\}$ with appropriate parity conditions).
3. **Welch**: $d = 2^t + 3$, $n = 2t + 1$. Conjectured by Welch, proved APN by Dobbertin [9].
4. **Niho**: $d = 2^t + 2^{t/2} - 1$ (even t) or $d = 2^t + 2^{(3t+1)/2} - 1$ (odd t), $n = 2t + 1$. Proved APN by Dobbertin, Helleseht, Kumar, and Martinsen [10].
5. **Inverse** [2]: $d = 2^n - 2$. APN only for n odd.
6. **Dobbertin** [11]: $d = 2^{4t} + 2^{3t} + 2^{2t} + 2^t - 1$, $n = 5t$.

No other infinite family of APN power maps is currently known. It is widely believed that this list is complete, but the conjecture remains open for $n > 26$.

5.5 Proof of the APN property for the Kasami exponent

The APN property of $x^{2^{2k}-2^k+1}$ was first established, implicitly, by Janwa and Wilson [5] using a Weil-bound argument on Fermat-variety intersections over $\mathbb{F}_2$. Their method counts rational points on certain algebraic curves to show that the number of solutions of the differential equation is at most 2.

An algebraic-combinatorial characterization was given by Carlet, Charpin, and Zinoviev [6], who expressed APN in terms of the associated cyclic code having minimum distance at least 5, and verified this for the Kasami exponent using cyclotomic coset arguments.

A self-contained elementary proof was given by Carlet, Kim, and Mesnager [7], whose paper *A direct proof of APN-ness of the Kasami functions* provides a direct algebraic argument without appeal to the algebraic-variety or coding-theory frameworks. This proof works for all $\gcd(k, n) = 1$ and in particular does not require n to be odd.

A third approach, closest to the one formalized in this project, proceeds through linearized polynomials and the decomposition of the differential map via the truncated trace L_k . This is the approach of Dempwolff and Müller [14], whose Theorem 3.2 provides the bijectivity of $x \mapsto L_k(x) \cdot x^{e'}$ that serves as the engine of the formal proof (Layer 6).

5.6 Role of the odd-dimension assumption

The APN property of the Kasami monomial $x^{2^{2k}-2^k+1}$ holds for *all* $\mathbb{F}_{2^n}$ satisfying $\gcd(k, n) = 1$, regardless of the parity of n . This is established, for instance, by the algebraic-variety argument of Janwa and Wilson [5] and by the cyclic-code criterion of Carlet, Charpin, and Zinoviev [6], neither of which requires n to be odd.

The odd-dimension assumption in the present formalization is therefore not an intrinsic property of the Kasami APN theorem, but rather a limitation of the particular proof method employed here. The Dempwolff–Müller approach [14] proceeds by constructing a linking exponent e' via the Gold coprimality identity $\gcd(2^k + 1, 2^n - 1) = 1$. That identity holds precisely when n is odd and $\gcd(k, n) = 1$; for even n the gcd can equal 2 or more, and the modular inverse of $2^k + 1$ modulo

$2^n - 1$ does not exist, making the construction of e' unavailable. Lifting the odd-dimension restriction would therefore require either a different proof strategy or an extension of the Dempwolff–Müller theorem to the even-dimensional case.

The condition k odd is an additional requirement of the Dempwolff–Müller bijectivity criterion for the linearized-polynomial component, and likewise does not reflect a necessary condition for the APN property itself.

5.7 Connection to Almost Bent functions

Over $\mathbb{F}_{2^n}$ with n odd, there is a strict hierarchy of nonlinearity properties. Chabaud and Vaude- nay [8] and Carlet, Charpin, and Zinoviev [6] established that every AB (Almost Bent) function is APN, but not conversely. The Kasami monomial is both APN and AB under the standard hypothe- ses. The AB property is characterized spectrally: all Walsh coefficients squared are in $\{0, 2^{n+1}\}$. The proof that Kasami is AB uses the APN result as an ingredient (via the fourth-moment identity) and then applies a general divisibility argument for character sums.

5.8 Walsh transform and moment identities

The Walsh-spectrum approach to APN and AB was systematized by Carlet, Charpin, and Zi- noviev [6]. A function is APN iff its fourth Walsh moment

$$\sum_{b \in F} W_f(a, b)^4 = 2|F|^3$$

holds for every nonzero a . This identity is a consequence of the differential-uniformity condition and gives a powerful spectral criterion. The moment approach allows APN and AB to be deduced from Parseval, fourth-moment, and divisibility conditions without requiring explicit knowledge of individual Walsh coefficients.

5.9 Computer search and classification results

Exhaustive computer searches have verified that no new APN power maps exist for small n . Berger et al. [12] confirmed that the six families in Section 6.3 account for all APN exponents up to $n = 26$. For general (non-power) APN functions, new sporadic examples over $\mathbb{F}_{2^6}$ were found [13], but for power maps the classification problem remains open.

5.10 Formal verification

The formalization in this project is, to the best of the authors' knowledge, the first complete machine-checked proof of the Kasami APN theorem in a proof assistant. The formal proof is structured around the Dempwolff–Müller linearized-polynomial approach [14] and makes the role of each hypothesis (n odd, k odd, $\gcd(k, n) = 1$) visible at the granularity of individual Lean lemmas.

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